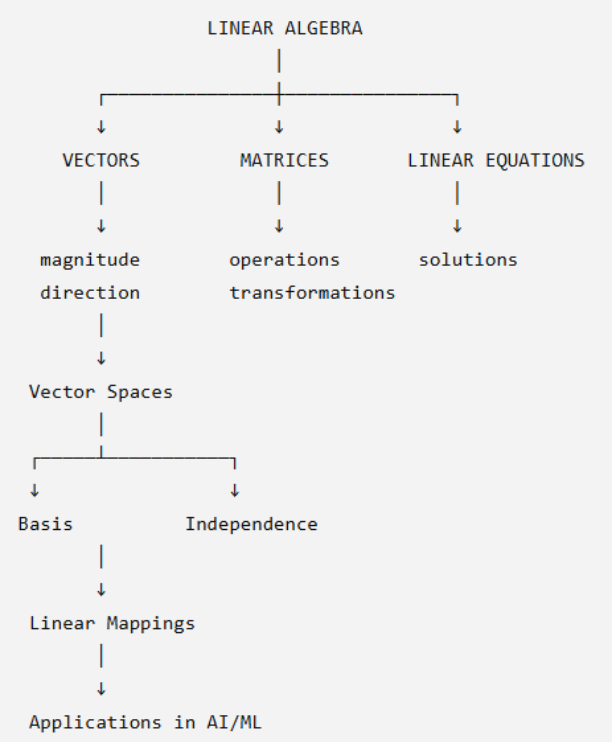
Mathematics for AI

DAY 1: Linear Algebra

"Linear algebra is the mathematics of representing and manipulating data using **vectors and matrices and linear equations**. Since computers represent large amounts of data numerically, linear algebra becomes one of the fundamental mathematical tools behind **AI and Machine Learning, Data Science, Computer Vision, Deep Learning.**



Why Linear Algebra is important in AI/ML

Almost every AI/ML dataset can be represented using vectors and matrices.

For example:

Dataset = Matrix

$$A = \begin{bmatrix} 80 & 75 & 90 \\ 70 & 85 & 80 \\ 90 & 95 & 88 \end{bmatrix}$$

One observation = Vector

$$\mathbf{x} = \begin{bmatrix} 80 \\ 75 \\ 90 \end{bmatrix}$$

Features = Vector components

Then machine-learning algorithms perform mathematical operations on these vectors and matrices.

Suppose I want to represent a house using three features: area, bedrooms and balconies. How can I represent it mathematically?

It can be represented as [1200, 3, 4] respectively which is a vector, understandable by AI.

AI/ML application	Linear Algebra concept
Dataset representation	Matrices
Feature representation	Vectors
Similarity	Dot product / Cosine similarity
PCA	Eigenvectors & eigenvalues
Neural networks	Matrix multiplication
Computer vision	Matrices / tensors
Recommendation systems	Matrix factorization
Embeddings	Vectors
Dimensionality reduction	Basis, rank, projections

Session 1: VECTORS

A quantity that involves only one value (magnitude) which is a real number and has no direction is called a **Scalar quantity**

A **Vector** is a mathematical quantity that has both **magnitude and direction**. It can be represented in a plane by a directed line segment.



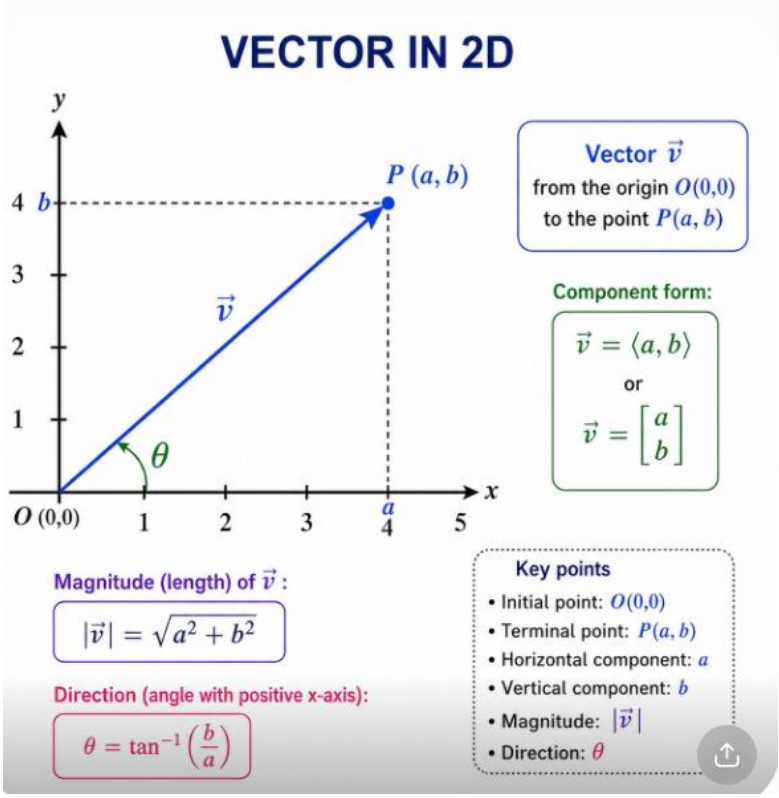
Ex: temperature = 30º, Weight = 60kg, Age = 20 years

In artificial intelligence, a vector is an **ordered list or array** of numbers that represents complex data like words, images, or audio so that a computer can process and understand it

Suppose we want to describe a student's marks: Marks $= (80, 75, 90)$ This is a **vector**.

Instead of writing three separate values, we can represent them together: X= $\begin{bmatrix} 80 \\ 75 \\ 90 \end{bmatrix}$

Two vectors are the same if they have the same magnitude and direction. Vectors are denoted as $\overrightarrow{AB}$ or $\vec{a}$ and the magnitude of the vector $\boldsymbol{a}$ by $\| \boldsymbol{a} \|$. The distance between initial and terminal points of a vector is called the **magnitude (or length)** of the vector, denoted as $|\overrightarrow{AB}|$, or $|a|$



Position Vectors in 3D plane

10.5.1 *Components of a vector*

Let us take the points A(1, 0, 0), B(0, 1, 0) and C(0, 0, 1) on the x-axis, y-axis and z-axis, respectively. Then, clearly

$$|\overrightarrow{OA}| = 1, |\overrightarrow{OB}| = 1 \text{ and } |\overrightarrow{OC}| = 1$$

The vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$, each having magnitude 1, are called *unit vectors along the axes* OX, OY and OZ, respectively, and denoted by $\hat{i}, \hat{j}$ and $\hat{k}$, respectively (Fig 10.13).

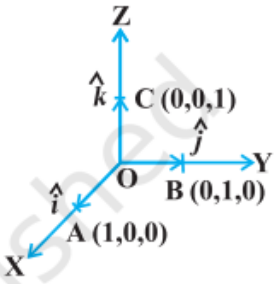


Fig 10.13

Now, consider the position vector $\overrightarrow{OP}$ of a point P(x, y, z) as in Fig 10.14. Let P_1 be the foot of the perpendicular from P on the plane XOY.

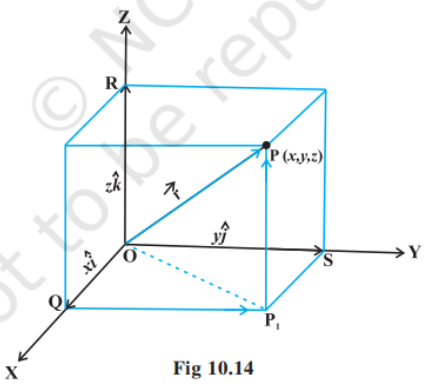


Fig 10.14

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Therefore, it follows that $\overrightarrow{OP_1} = \overrightarrow{OQ} + \overrightarrow{QP_1} = x\hat{i} + y\hat{j}$

and $\overrightarrow{OP} = \overrightarrow{OP_1} + \overrightarrow{P_1P} = x\hat{i} + y\hat{j} + z\hat{k}$

Hence, the position vector of P with reference to O is given by

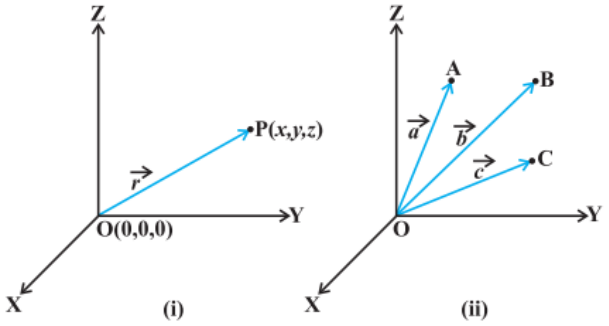
$$\overrightarrow{OP} \text{ (or } \vec{r}) = x\hat{i} + y\hat{j} + z\hat{k}$$

This form of any vector is called its *component form*. Here, x, y and z are called as the *scalar components* of $\vec{r}$, and $x\hat{i}$, $y\hat{j}$ and $z\hat{k}$ are called the *vector components* of $\vec{r}$ along the respective axes. Sometimes x, y and z are also termed as *rectangular components*.

Hence, the length of any vector $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ is given by

$$|\vec{r}| = |x\hat{i} + y\hat{j} + z\hat{k}| = \sqrt{x^2 + y^2 + z^2}$$

In practice, the position vectors of points A, B etc., with respect to the origin O are denoted by $\overline{a}$, $\overline{b}$ etc., respectively



Direction Cosines

Direction cosines of a vector are the cosines of the angles that the vector makes with the positive axes of a coordinate system.

Consider the position vector $\overrightarrow{OP}$ (or $\vec{r}$) of a point P(x, y, z) as in Fig 10.3. The angles α, β, γ made by the vector $\vec{r}$ with the positive directions of x, y and z-axes respectively, are called its *direction angles*. The cosine values of these angles, i.e., $\cos\alpha$, $\cos\beta$ and $\cos\gamma$ are called *direction cosines* of the vector $\vec{r}$, and usually denoted by l, m and n,

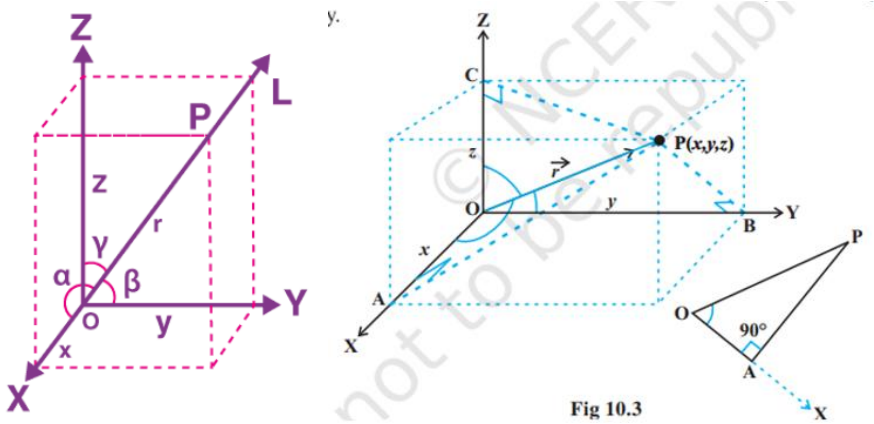
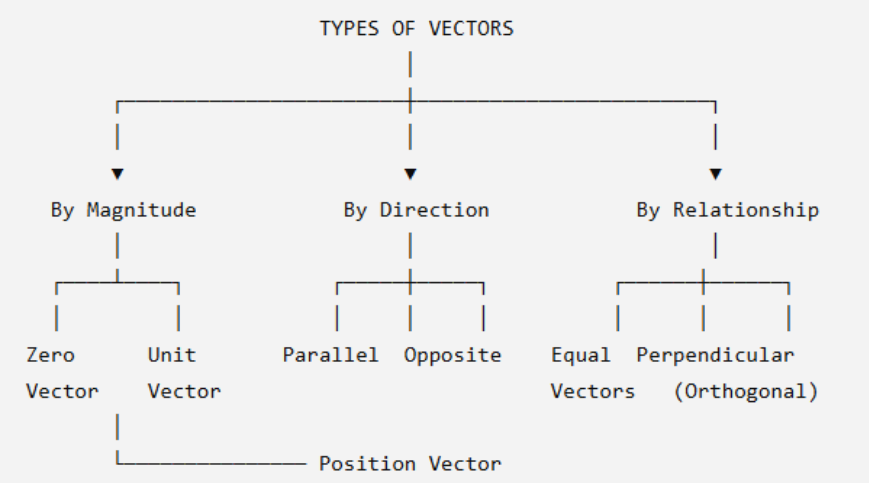


Fig 10.3

From Fig 10.3, one may note that the triangle OAP is right angled, and in it, we have $\cos\alpha = \frac{x}{r}$ (r stands for $|\vec{r}|$). Similarly, from the right angled triangles OBP and OCP, we may write $\cos\beta = \frac{y}{r}$ and $\cos\gamma = \frac{z}{r}$. Thus, the coordinates of the point P may Expressed as (lr, mr, nr).

One may note that $l^2 + m^2 + n^2 = 1$ but $a^2 + b^2 + c^2 \neq 1$, in general.

TYPES OF VECTORS



Zero Vector A vector whose initial and terminal points coincide, is called a zero vector (or null vector), and denoted as $\vec{0}$. Zero vector can not be assigned a definite direction as it has zero magnitude. Or, alternatively otherwise, it may be regarded as having any direction. The vectors $\overrightarrow{AA}$, $\overrightarrow{BB}$ represent the zero vector,

Unit Vector A vector whose magnitude is unity (i.e., 1 unit) is called a unit vector. The unit vector in the direction of a given vector $\vec{a}$ is denoted by $\hat{a}$.

$$\hat{a} = \frac{1}{|\vec{a}|}\vec{a}$$

Coinitial Vectors Two or more vectors having the same initial point are called coinitial vectors.

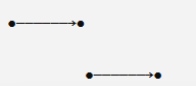
Collinear Vectors Two or more vectors are said to be collinear if they are parallel to the same line, irrespective of their magnitudes and directions.



Negative of a Vector A vector whose magnitude is the same as that of a given vector (say, $\overrightarrow{AB}$), but direction is opposite to that of it, is called *negative* of the given vector. For example, vector $\overrightarrow{BA}$ is negative of the vector $\overrightarrow{AB}$, and written as $\overrightarrow{BA} = -\overrightarrow{AB}$.

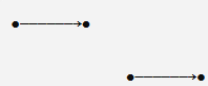


Equal Vectors Two vectors $\vec{a}$ and $\vec{b}$ are said to be equal, if they have the same magnitude and direction regardless of the positions of their initial points, and written as $\vec{a} = \vec{b}$.



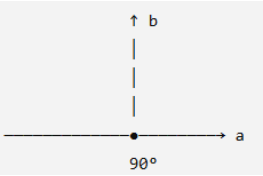
Like Vectors: Two vectors are said to be like vectors, if they have same direction and they are unlike if they are in opposite directions.

Free Vector: A Vector is said to be free or non-localised vector, if its origin can be taken anywhere in space.

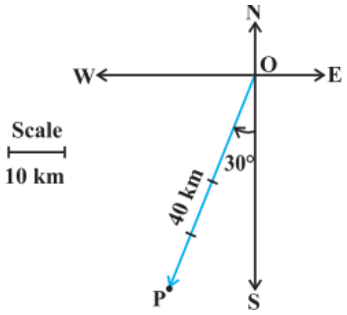


Orthogonal Vectors : Two vectors are perpendicular / orthogonal if they meet at 90º. Their dot product is 0.

$$\vec{a} \cdot \vec{b} = 0$$



Ex: Represent graphically a displacement of 40km, 30º west of south.



Ex: In the given figure, find Collinear vectors, Equal vectors, Coinitial vectors

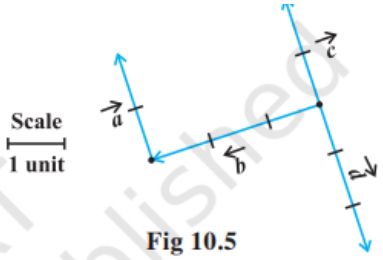


Fig 10.5

(i) Collinear vectors : $\vec{a}$, $\vec{c}$ and $\vec{d}$.

(ii) Equal vectors : $\vec{a}$ and $\vec{c}$.

(iii) Coinitial vectors : $\vec{b}$, $\vec{c}$ and $\vec{d}$.

10.4 Addition of Vectors

A vector $\overrightarrow{AB}$ simply means the displacement from a point A to the point B. Now consider a situation that a girl moves from A to B and then from B to C (Fig 10.7). The net displacement made by the girl from point A to the point C, is given by the vector $\overrightarrow{AC}$ and expressed as

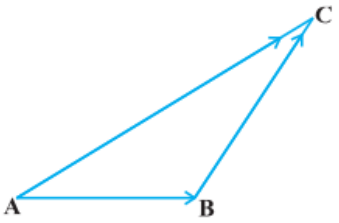
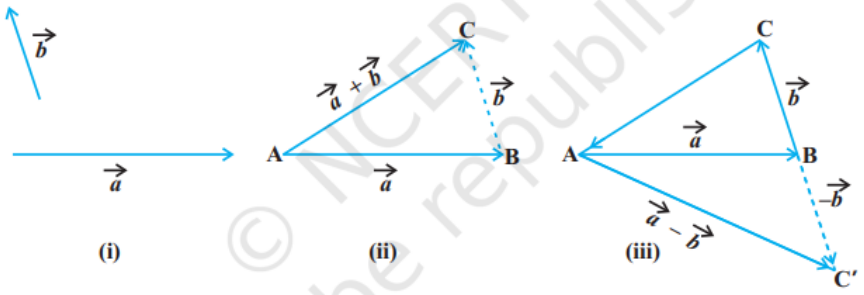


Fig 10.7

$$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC}$$

This is known as the *triangle law of vector addition.*

If we have two vectors , then to add them, they are positioned so that the initial point of one coincides with the terminal point of the other.



$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$$

Now again, since $\overrightarrow{AC} = -\overrightarrow{CA}$, from the above equation, we have

$$\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CA} = \overrightarrow{AA} = \vec{0}$$

Now, construct a vector $\overrightarrow{BC'}$ so that its magnitude is same as the vector $\overrightarrow{BC}$, but the direction opposite to that of it (Fig 10.8 (iii)), i.e.,

$$\overrightarrow{BC'} = -\overrightarrow{BC}$$

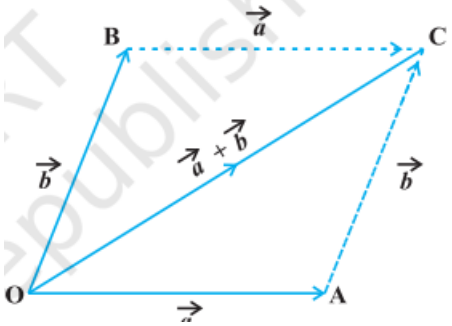
Then, on applying triangle law from the Fig 10.8 (iii), we have

$$\overrightarrow{AC'} = \overrightarrow{AB} + \overrightarrow{BC'} = \overrightarrow{AB} + (-\overrightarrow{BC}) = \vec{a} - \vec{b}$$

The vector $\overrightarrow{AC'}$ is said to represent the *difference of* $\vec{a}$ and $\vec{b}$.

Now consider a boat in a river going from one bank of the river to the other in a direction perpendicular to the flow of the river. Then it is acted upon by two velocity vectors – one is the velocity imparted to the boat by its engine and the other is the velocity of the flow of river water. Under the influence of these two velocities, the boat starts travelling with a different velocity.

If we have two vectors $\vec{a}$ and $\vec{b}$ represented by the two adjacent sides of a parallelogram in magnitude and direction (Fig 10.9), then their sum $\vec{a} + \vec{b}$ is represented in magnitude and direction by the diagonal of the parallelogram through their common point. This is known as the *parallelogram law of vector addition.*



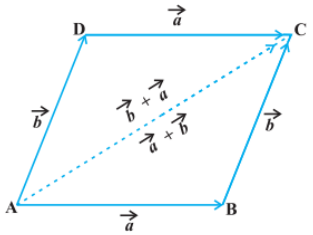
$\overrightarrow{OA} + \overrightarrow{AC} = \overrightarrow{OC}$

or $\overrightarrow{OA} + \overrightarrow{OB} = \overrightarrow{OC}$ (since $\overrightarrow{AC} = \overrightarrow{OB}$)

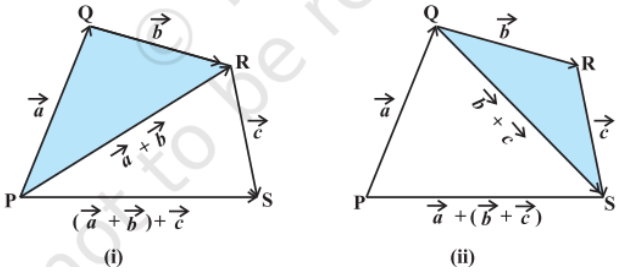
which is parallelogram law. Thus, we may say that the two laws of vector addition are equivalent to each other.

Properties of Vector Addition

1. $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ (Commutative property)



2. $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$ (Associative property)



3. $\vec{a} + \vec{0} = \vec{0} + \vec{a} = \vec{a}$ (Additive identity)

Here, the zero vector $\vec{0}$ is called the *additive identity* for the vector addition.

Why Vector Addition is Important in AI/ML?

Vectors represent data and features in machine learning. For example:

$$\text{Student}_1 = (80, 70, 90)$$
$$\text{Student}_2 = (75, 85, 80)$$

Vector addition = component-wise addition.

$$(a_1, a_2) + (b_1, b_2) = (a_1 + b_1,\ a_2 + b_2)$$

Vector operations allow algorithms to manipulate these feature representations.

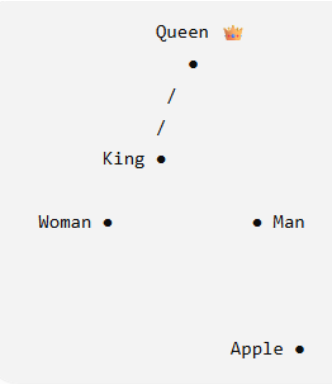
Vector addition is also used in:

- Neural networks: Neural networks combine different inputs using **weights** and addition.

$$Output = w_1x_1 + w_2x_2 + b$$

- Word embeddings: AI converts words into vectors. Words are represented as numbers, and vector operations help AI understand relationships between words.

Word	Example vector
King	`[0.52, 0.81, 0.12, 0.67]`
Queen	`[0.50, 0.79, 0.15, 0.70]`
Man	`[0.48, 0.72, 0.10, 0.61]`
Woman	`[0.46, 0.70, 0.14, 0.65]`
Apple	`[0.12, 0.25, 0.91, 0.18]`



Words with similar meanings tend to have vectors that are **close together** in the embedding space.

- Computer vision: Images can be represented using numbers/vectors.

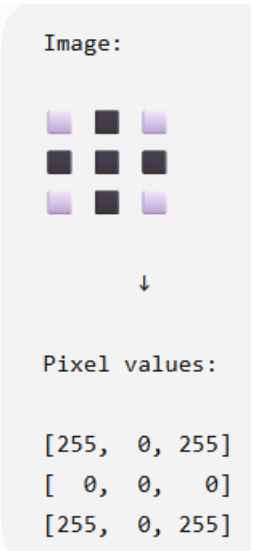
  In **Computer Vision**, vectors are used to convert an image—which is visual information—into **numbers that a computer can process**.

  A simple way to explain it to students is:

  Image → Pixels → Numbers → Vectors → AI understands the image

For a grayscale image, each pixel can have a value from **0 to 255**:

- `0` → black
- `255` → white
- values in between → shades of gray



- Recommendation systems: Vector operations are part of how these representations are processed.
- Optimization: means **finding the best solution**.

  AI models use vectors to represent parameters. For example: $W = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$

  During training, the model updates these values:

  $$W_{new} = W + \Delta W$$

  So vector addition is used to **update model parameters** during learning.
- Feature representation: This is one of the **most important uses in Machine Learning**.

  Suppose a house is represented by:

  $House = \begin{bmatrix} 2000 \\ 3 \\ 2 \end{bmatrix}$ where: 2000 = area, 3 = bedrooms, 2 = bathrooms

  Another set of information can be represented as another vector.

  AI combines different features to create a richer representation of the house.

Multiplication of a vector by a scalar

If $\bar{a}$ is a vector and λ is a scalar, then $\lambda\bar{a}$ is called multiplication of vector by scalar. $\lambda\bar{a}$ is a vector with direction same or opposite depending on value of λ being positive or negative. Also its magnitude is λ times the magnitude of vector $\bar{a}$.

$$|\lambda\vec{a}| = |\lambda||\vec{a}|$$

If $\vec{a}$ and $\vec{b}$ are any two vectors given in the component form $a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, respectively, then

(i) the sum (or resultant) of the vectors $\vec{a}$ and $\vec{b}$ is given by

$$\vec{a} + \vec{b} = (a_1 + b_1)\hat{i} + (a_2 + b_2)\hat{j} + (a_3 + b_3)\hat{k}$$

(ii) the difference of the vector $\vec{a}$ and $\vec{b}$ is given by

$$\vec{a} - \vec{b} = (a_1 - b_1)\hat{i} + (a_2 - b_2)\hat{j} + (a_3 - b_3)\hat{k}$$

(iii) the vectors $\vec{a}$ and $\vec{b}$ are equal if and only if

$$a_1 = b_1,\ a_2 = b_2 \quad \text{and} \quad a_3 = b_3$$

(iv) the multiplication of vector $\vec{a}$ by any scalar λ is given by

$$\lambda\vec{a} = (\lambda a_1)\hat{i} + (\lambda a_2)\hat{j} + (\lambda a_3)\hat{k}$$

Two vectors $\bar{a}$, $\bar{b}$ are collinear if there exists a nonzero scalar λ such that $\vec{b} = \lambda\vec{a}$.

in the component form, i.e. $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, then the two vectors are collinear if and only if

$$b_1\hat{i} + b_2\hat{j} + b_3\hat{k} = \lambda(a_1\hat{i} + a_2\hat{j} + a_3\hat{k})$$

$$\Leftrightarrow \quad b_1\hat{i} + b_2\hat{j} + b_3\hat{k} = (\lambda a_1)\hat{i} + (\lambda a_2)\hat{j} + (\lambda a_3)\hat{k}$$

$$\Leftrightarrow \quad b_1 = \lambda a_1,\ b_2 = \lambda a_2,\ b_3 = \lambda a_3$$

$$\Leftrightarrow \quad \frac{b_1}{a_1} = \frac{b_2}{a_2} = \frac{b_3}{a_3} = \lambda$$

10.5.2 *Vector joining two points*

If $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ are any two points, then the vector joining P_1 and P_2 is the vector $\overrightarrow{P_1P_2}$ (Fig 10.15).

Joining the points P_1 and P_2 with the origin O, and applying triangle law, from the triangle OP_1P_2, we have

$$\overrightarrow{OP_1} + \overrightarrow{P_1P_2} = \overrightarrow{OP_2}$$

Using the properties of vector addition, the above equation becomes

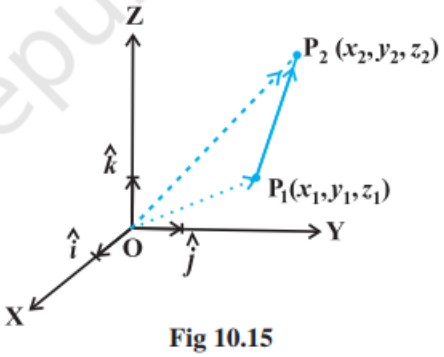


Fig 10.15

$$\overrightarrow{P_1P_2} = \overrightarrow{OP_2} - \overrightarrow{OP_1}$$

i.e. $$\overrightarrow{P_1P_2} = (x_2\hat{i} + y_2\hat{j} + z_2\hat{k}) - (x_1\hat{i} + y_1\hat{j} + z_1\hat{k})$$

$$= (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$$

The magnitude of vector $\overrightarrow{P_1P_2}$ is given by

$$|\overrightarrow{P_1P_2}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

PRODUCT OF TWO VECTORS

Scalar (or dot) product of two vectors

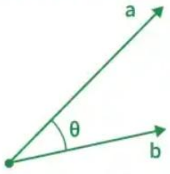
The dot product (or scalar product) of two vectors is a way of multiplying vectors that results in a **scalar (number).**

$$a \cdot b = \sum_{i=1}^{n} a_i b_i$$

$$a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \qquad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$$a.b = a_1b_1 + a_2b_2 + \ldots a_nb_n$$

$$a \cdot b = |a|\,|b| \cos(\theta)$$



$$|a| = \sqrt{a_1^2 + a_2^2 + \ldots + a_n^2}$$

$$|b| = \sqrt{b_1^2 + b_2^2 + \ldots + b_n^2}$$

$$\begin{bmatrix} 2 \\ 7 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 8 \\ 2 \\ 8 \end{bmatrix}$$

Dot product

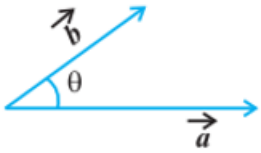
$2 \cdot 8$

$7 \cdot 2$

$1 \cdot 8$

defined as $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|\cos\theta$,

where, θ is the angle between $\vec{a}$ and $\vec{b}$, $0 \le \theta \le \pi$ (Fig 10.19).



$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$$

Property 1 (Distributivity of scalar product over addition) Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be any three vectors, then

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

Property 2 Let $\vec{a}$ and $\vec{b}$ be any two vectors, and l be any scalar. Then

$$(\lambda\vec{a}) \cdot \vec{b} = (\lambda\vec{a}) \cdot \vec{b} = \lambda(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (\lambda\vec{b})$$

Core Uses of Dot Product in AI

- **Calculations in Artificial Neurons:** Every single neuron in a neural network performs a weighted sum of its inputs. This operation is a direct dot product between the input vector (x) and the weight vector (w), plus a bias term ($h = w \cdot x + b$).

```
Input                Weight
 x₁ = 2   ————————→  w₁ = 0.5
 x₂ = 3   ————————→  w₂ = 0.2
 x₃ = 4   ————————→  w₃ = 0.1

              ↓
       Dot Product

(2×0.5)+(3×0.2)+(4×0.1)
              ↓
            2.0
              ↓
       Activation Function
              ↓
           Output
```

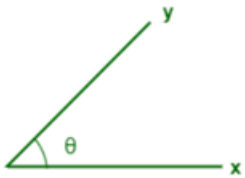
- **Measuring Vector Similarity:** The dot product yields a larger positive value when two vectors point in similar directions. AI models use this property—often via Cosine Similarity—to determine if two images, documents, or concepts are related.

```
User Vector       Product Vector
     ↘                 ↙
       \             /
        \ Dot Product /
         \         /
          ↓       ↓
        Similarity Score
              ↓
       Recommendation
```

- **Recommendation Systems:** Platforms use embedding vectors to represent user preferences and item features. Taking the dot product of a user vector and an item vector calculates a match score to recommend videos, products, or movies.
- **Attention Mechanisms in Transformers:** Large language models (LLMs) use dot products inside attention layers to score how much focus one word in a sentence should place on other words, establishing contextual alignment.

Cosine Similarity

Cosine similarity is a metric used to measure how **similar two vectors** are by calculating the cosine of the angle between them. It measures the orientation or direction of the vectors in a multi-dimensional space, ignoring their size or length.

Cosine similarity = $\frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$, 

Cosine similarity b/w two vectors

The cosine similarity between two vectors is measured in 'θ'.

Smaller the angle, higher the similarity.

Cosine Similarity ranges from -1 to +1.

Score Interpretation

- **Score of 1:** The vectors point in the exact same direction (angle is 0°), meaning they are identical in context.
- **Score of 0:** The vectors are orthogonal or perpendicular (angle is 90°), meaning they are completely unrelated.
- **Score of -1:** The vectors point in opposite directions (angle is 180°), meaning they are exact opposites.

Common Uses: It is widely used in machine learning and data analysis :

- **Text analysis:** Comparing two documents or search queries based on word frequencies, regardless of document length.
- **Recommendation systems:** Measuring user preference overlap for items like movies or products.
- **Machine learning:** Comparing embeddings in natural language processing and computer vision.

Projection of a vector on a line

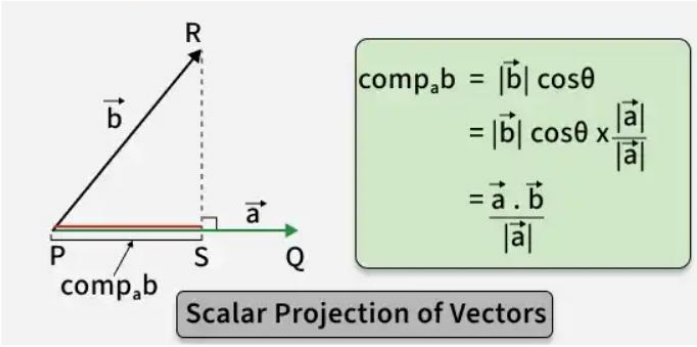
The projection of a vector on a line is the shadow or component of that vector that lies along the direction of the line.

Scalar projection gives the length (or magnitude) of that shadow. It can be positive or negative.

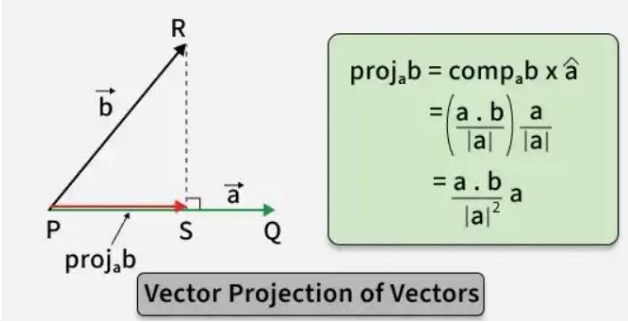
Vector projection gives both the length and the direction of the shadow along the line.

Scalar projection tells us how much of a lies in the direction of b, Vector projection gives the actual vector .

1. Scaler Projection of Vector



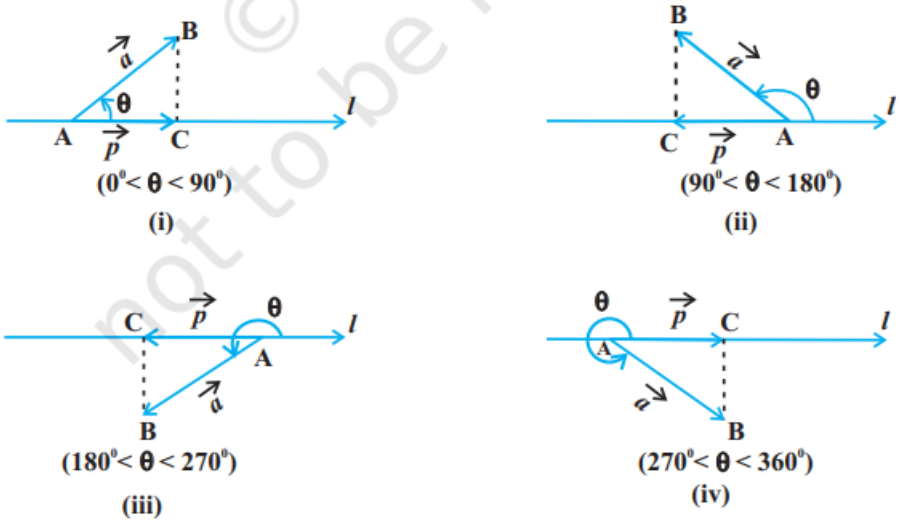
2. Vector Projection of Vector



$$\text{comp}_{\mathbf{b}}(\mathbf{a}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|} \quad , \quad \text{proj}_{\mathbf{b}}(\mathbf{a}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|^2}\mathbf{b}$$

Suppose a vector $\overrightarrow{AB}$ makes an angle θ with a given directed line l (say), in the *anticlockwise direction* (Fig 10.20). Then the projection of $\overrightarrow{AB}$ on l is a vector $\vec{p}$ (say) with magnitude $|\overrightarrow{AB}|\,|\cos\theta|$, and the direction of $\vec{p}$ being the same (or opposite) to that of the line l, depending upon whether $\cos\theta$ is positive or negative. The vector $\vec{p}$

is called the *projection vector*, and its magnitude $|\vec{p}|$ is simply called as the *projection* of the vector $\overrightarrow{AB}$ on the directed line l.



Observations

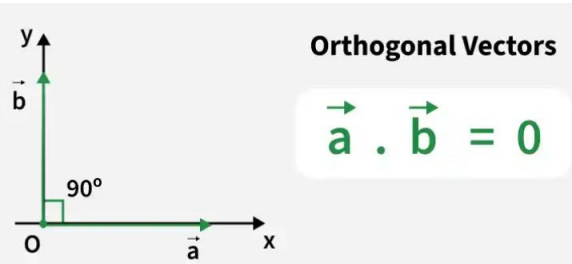
1. If $\hat{p}$ is the unit vector along a line l, then the projection of a vector $\vec{a}$ on the line l is given by $\vec{a} \cdot \hat{p}$.
2. Projection of a vector $\vec{a}$ on other vector $\vec{b}$, is given by

$$\vec{a} \cdot \hat{b}, \quad \text{or} \quad \vec{a} \cdot \left(\frac{\vec{b}}{|\vec{b}|} \right), \quad \text{or} \quad \frac{1}{|\vec{b}|} (\vec{a} \cdot \vec{b})$$

3. If $\theta = 0$, then the projection vector of $\overline{AB}$ will be $\overline{AB}$ itself and if $\theta = \pi$, then the projection vector of $\overline{AB}$ will be $\overline{BA}$.
4. If $\theta = \frac{\pi}{2}$ or $\theta = \frac{3\pi}{2}$, then the projection vector of $\overline{AB}$ will be zero vector.

Orthogonal Vectors

Orthogonal vectors are vectors that are perpendicular to each other and form an angle of 90°. In vector algebra, two vectors are said to be orthogonal when their dot product is zero. Orthogonality is the property that describes the perpendicular relationship between two vectors.



Properties

Various properties of Orthogonal vectors are:

- **Perpendicular Nature**: Orthogonal vectors are always perpendicular to each other.
- **Dot Product**: The dot product of orthogonal vectors is zero.

- **Null Vector**: A null vector (a vector with zero magnitude) is orthogonal to every vector.

Orthogonal Projection of a Vector

"Orthogonal projection is the process of finding the component of one vector along another vector, with the remaining component perpendicular to the target vector."

Suppose we have two vectors:

a= vector we want to project

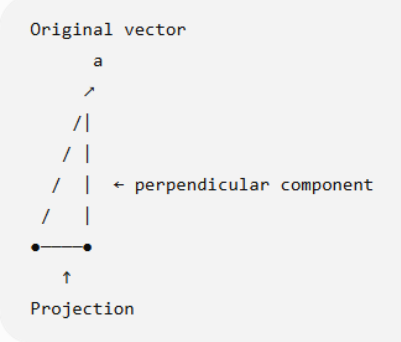
b= direction onto which we project

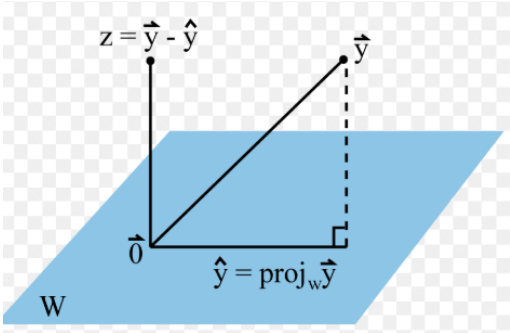
So orthogonal projection of a onto b is the shadow or component of a that lies directly along the direction b

Why is it called orthogonal projection?

Because the part removed from the original vector is **perpendicular (90°)** to the target vector.

$$\mathbf{a} = \underbrace{proj_{\mathbf{b}}(\mathbf{a})}_{\text{parallel component}} + \underbrace{\mathbf{r}}_{\text{perpendicular component}}$$





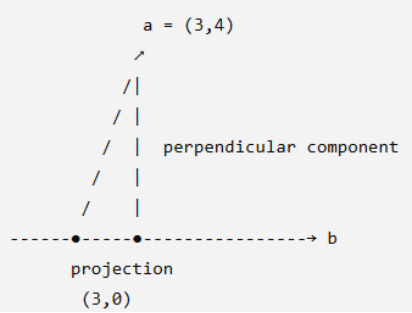
Consider two vectors a and b, the orthogonal projection of "a" onto "b", is expressed as:

$$\text{proj}_{\vec{b}}\vec{a} = \left(\frac{\vec{a}\cdot\vec{b}}{|\vec{b}|^2}\right)\vec{b}$$

Let $a = \begin{bmatrix}3\\4\end{bmatrix}$ and $b = \begin{bmatrix}1\\0\end{bmatrix}$

$$a \cdot b = (3)(1) + (4)(0) = 3$$
$$\| b \|^2 = 1^2 + 0^2 = 1$$
$$\text{proj}_b(a) = \frac{3}{1}\begin{bmatrix}1\\0\end{bmatrix} = \boxed{\text{proj}_b(a) = \begin{bmatrix}3\\0\end{bmatrix}}$$

So the vector $\begin{bmatrix}3\\4\end{bmatrix}$ has **parallel component:** $[3, 0]$ and **perpendicular component:** $[0, 4]$



⭐ Why is vector projection useful in AI?

AI Area	Use of Projection
Machine Learning	Reduce data to important directions
PCA	Project high-dimensional data onto principal components
Computer Vision	Project 3D information onto 2D images
NLP	Project word representations into useful feature spaces
Optimization	Find components along a particular direction
Signal Processing	Separate a signal into meaningful components
Neural Networks	Transform vectors into different representation spaces

Most important AI example: PCA

Suppose each customer has 10 features:

$$[x_1, x_2, \ldots, x_{10}]$$

Instead of using all 10 dimensions, PCA finds important directions and **projects the data onto those directions**.

```
10-dimensional data
        ↓
Find important directions
        ↓
Principal Components
        ↓
Project data
        ↓
2-dimensional representation
        ↓
Visualization / ML model
```

Vector (or cross) product of two vectors

Definition 3 The vector product of two nonzero vectors $\vec{a}$ and $\vec{b}$, is denoted by $\vec{a} \times \vec{b}$ and defined as

$$\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin\theta \, \hat{n},$$

where, θ is the angle between $\vec{a}$ and $\vec{b}$, $0 \le \theta \le \pi$ and $\hat{n}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$, such that $\vec{a}, \vec{b}$ and $\hat{n}$ form a right handed system (Fig 10.23). i.e., the right handed system rotated from $\vec{a}$ to $\vec{b}$ moves in the direction of $\hat{n}$.

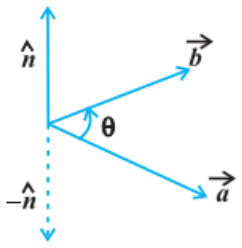
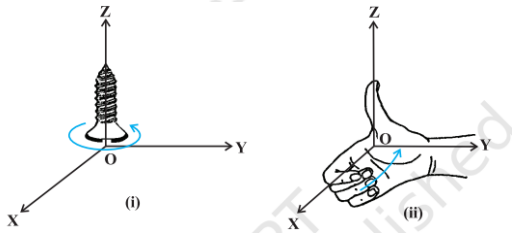


Fig 10.23



Observations

1. $\vec{a} \times \vec{b}$ is a vector.
2. Let $\vec{a}$ and $\vec{b}$ be two nonzero vectors. Then $\vec{a} \times \vec{b} = \vec{0}$ if and only if $\vec{a}$ and $\vec{b}$ are parallel (or collinear) to each other, i.e.,

$$\vec{a} \times \vec{b} = \vec{0} \Leftrightarrow \vec{a} \parallel \vec{b}$$

   In particular, $\vec{a} \times \vec{a} = \vec{0}$ and $\vec{a} \times (-\vec{a}) = \vec{0}$, since in the first situation, $\theta = 0$ and in the second one, $\theta = \pi$, making the value of $\sin\theta$ to be 0.

3. If $\theta = \frac{\pi}{2}$ then $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|$.
4. In view of the Observations 2 and 3, for mutually perpendicular unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$ (Fig 10.24), we have

$$\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{0}$$
$$\hat{i} \times \hat{j} = \hat{k}, \quad \hat{j} \times \hat{k} = \hat{i}, \quad \hat{k} \times \hat{i} = \hat{j}$$



Fig 10.24

5. In terms of vector product, the angle between two vectors $\vec{a}$ and $\vec{b}$ may be given as

$$\sin\theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}$$

6. It is always true that the vector product is not commutative, as $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$.

Hence $\vec{a}\times\vec{b} = |\vec{a}||\vec{b}|\sin\theta\hat{n}$

$= -|\vec{a}||\vec{b}|\sin\theta\hat{n}_1 = -\vec{b}\times\vec{a}$

7. In view of the Observations 4 and 6, we have

$$\hat{j}\times\hat{i} = -\hat{k},\quad \hat{k}\times\hat{j} = -\hat{i} \text{ and } \hat{i}\times\hat{k} = -\hat{j}.$$

8. If $\vec{a}$ and $\vec{b}$ represent the adjacent sides of a triangle then its area $\frac{1}{2}|\vec{a}\times\vec{b}|$.

C

9. If $\vec{a}$ and $\vec{b}$ represent the adjacent sides of a parallelogram, then its area is given

Area of parallelogram ABCD $= |\vec{b}\,||\vec{a}|\sin\theta = |\vec{a}\times\vec{b}|$.

Property 3 (Distributivity of vector product over addition): If $\vec{a}, \vec{b}$ and $\vec{c}$ are any three vectors and λ be a scalar, then

(i) $\vec{a}\times(\vec{b}+\vec{c}) = \vec{a}\times\vec{b} + \vec{a}\times\vec{c}$

(ii) $\lambda(\vec{a}\times\vec{b}) = (\lambda\vec{a})\times\vec{b} = \vec{a}\times(\lambda\vec{b})$

Let $\vec{a}$ and $\vec{b}$ be two vectors given in component form as $a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ and $b_1\hat{i}+b_2\hat{j}+b_3\hat{k}$, respectively. Then their cross product may be given by

$$\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Normal and Orthonormal Vectors in Linear algebra

In **Linear Algebra for AI**, we often need to know not only the direction of a vector but also its **length (magnitude)**. This leads to the concepts of **normal vectors** and **orthonormal vectors**.

A vector is called a **normal (or normalized) vector** when its **magnitude is 1**.

If

$$v = (v_1, v_2, \ldots, v_n)$$

then its magnitude is $\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$

To convert a non-zero vector into a unit/normalized vector:

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$$

This process is called **normalization**.

Consider v = (3, 4) then magnitude is $\|\mathbf{v}\| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = 5$ and Normalised vector will be

$$\hat{\mathbf{v}} = \frac{(3,4)}{5}$$

$$\boxed{\hat{\mathbf{v}} = \left(\frac{3}{5}, \frac{4}{5}\right)}$$ if $\|\hat{\mathbf{v}}\| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = 1$ then $\left(\frac{3}{5}, \frac{4}{5}\right)$ is a **unit/normalized vector**.

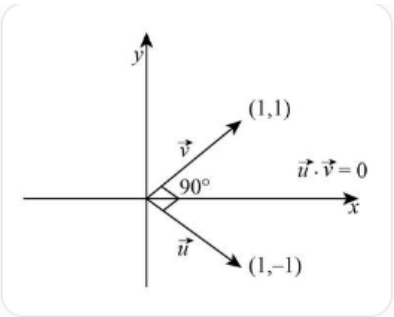
Two vectors are said to be **orthogonal** if they are perpendicular to each other (dot product is 0) in the same coordinate system. In three-dimensional space, this means they lie on mutually perpendicular planes.

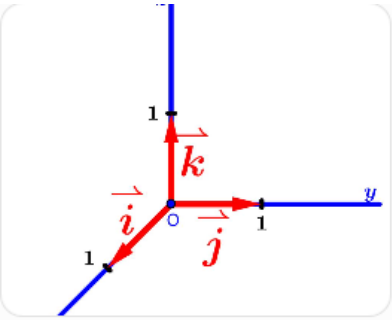
Ex: Consider two vectors: u= (3, 4) and v= (−4, 3).

Their dot product is: (3 × −4)+ (4 × 3)= −12+12=0

Since the dot product is zero, u and v are **orthogonal**.

Vectors which are orthogonal to each other and individually have unit magnitude (= 1) are known as **Orthonormal Vectors**.





"Orthonormal vectors give AI algorithms a clean set of independent perpendicular, unit-length directions. This makes data representation, projection, dimensionality reduction, matrix calculations, and numerical computation simpler and more stable."

Note: All orthonormal vectors are orthogonal by the definition itself.

.

Concept	Meaning	Condition
Normalized / Unit vector	Length is 1	$\|\mathbf{v}\| = 1$
Orthogonal vectors	Perpendicular to each other	$\mathbf{a} \cdot \mathbf{b} = 0$
Orthonormal vectors	Perpendicular **and** unit length	Dot product = 0 and magnitude = 1

One of the biggest applications of Orthonormal vectors is **Principal Component Analysis (PCA).

For a data with many features, PCA tries to transform data into new directions called **principal components**, which reduces many features into fewer important dimensions as PC_1, PC_2, PC_3,…which are mutually orthogonal principal directions and have magnitude 1.

Orthogonality helps PCA avoid repeating the same information. Unit magnitude helps PCA to focus on the direction and the amount of variation rather than dealing with arbitrary lengths of the direction vectors.

This helps with:

- Dimensionality reduction
- Removing redundancy
- Visualization
- Noise reduction
- Faster ML models

Orthonormality is also used in **neural-network weight initialization and training techniques.

If weights become poorly scaled, signals moving through many layers can:

- become extremely small → **vanishing gradients**
- become extremely large → **exploding gradients**

Orthogonal or orthonormal weight matrices can help maintain a reasonable scale of signals and gradients, particularly in certain deep or recurrent architectures.

** Orthonormal vectors are used in **Computer Vision**

Images can be represented as vectors or matrices. Transformations such as:

- rotation
- coordinate changes
- image representations can use orthonormal matrices.

Orthonormal vectors are particularly useful in:

- **PCA (Principal Component Analysis)**
- **Feature transformations**
- **Dimensionality reduction**
- **Computer graphics**
- **Machine learning**
- **Coordinate transformations**

Vector Norm and Vector Space

A **vector norm** is a mathematical function used to measure the **size or length of a vector**.

We write norm of vector as: $\|\mathbf{v}\|$

If $x=(x_1, x_2, x_3, \ldots, x_n)$ then $\sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$ is the length of the vector.

For vector x = (3, 4) then $\sqrt{3^2 + 4^2}$ = 5 which is the length of the vector.

Types of Vector Norms:

A. L1 Norm — Manhattan Norm: $\|\mathbf{x}\|_1 = |x_1| + |x_2| + \cdots + |x_n|$

For x = (3, -4) $\|\mathbf{x}\|_1 = |3| + |-4| = 3 + 4 = \boxed{7}$

AI/ML use: L1 norm is strongly associated with **Lasso regression** and sparse solutions.

B. L2 Norm — Euclidean Norm: $\|\mathbf{x}\|_2 = \sqrt{\sum x_i^2}$

For x = (3,4) $\|\mathbf{x}\|_2 = 5$

AI/ML use: Distance calculations, optimization, regularization and many machine-learning algorithms.

C. L∞ Norm — (Maximum Norm): $\|\mathbf{x}\|_\infty = \max_i |x_i|$

For x = (3, -4, 2) $\|\mathbf{x}\|_\infty = \max(3, 4, 2) = \boxed{4}$

VECTOR SPACE:

A vector space (also called linear space) is a mathematical structure formed by a collection of elements called vectors that can be **added together** and **scaled** by numbers called scalars, following specific rules known as axioms. Vector spaces form the foundation of linear algebra.

A **vector space** is a collection of vectors where we can perform:

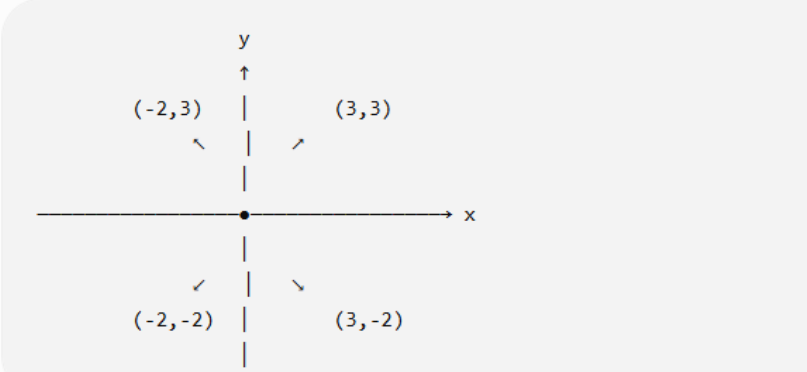
1. **Vector addition**
2. **Scalar multiplication** and the result remain within the same collection.

   For a vector a = (2, 3) , b = (4, 1)

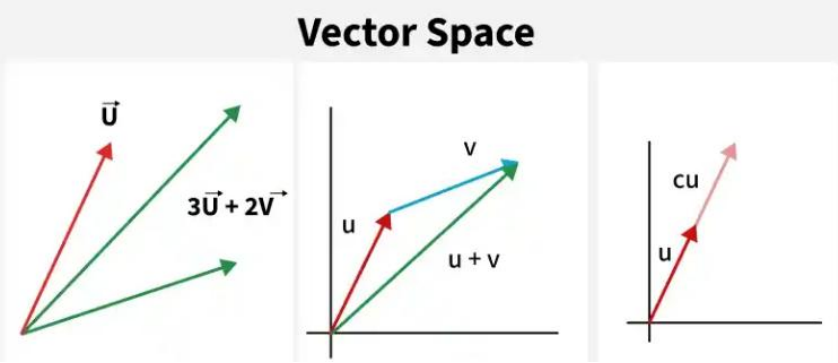
   a + b = (2 +4, 3 + 1) = (6, 4) and the result is a 2D vector

   3a = 3(2, 3) = (6, 9) and this result is a 2D vector.

We can visualize $\mathbb{R}^2$ as the entire 2D plane:



Every point/vector in this plane belongs to the vector space $\mathbb{R}^2$.



Linear Combination and Linear span

A **linear combination** means **multiplying vectors by scalars and then adding the results.**

Suppose we have two vectors: $u = (u_1, u_2)$ and $v = (v_1, v_2)$
A linear combination of these vectors is: $\boxed{w = au + bv}$ where a and b are **scalars** (numbers).

Example:

Let $u = (2, 1)$ and $v = (1, 3)$. If we take scalars as $a = 2,\ b = 3$ then,

$$w = 2u + 3v$$

so $2u = 2(2, 1) = (4, 2)$ and $3v = 3(1, 3) = (3, 9)$

therefore $w = (4, 2) + (3, 9) = (7, 11)$

So **(7,11) is a linear combination of (2,1) and (1,3).**

LINEAR SPAN

The **linear span** of a set of vectors is the **set of all possible linear combinations** that can be formed from those vectors.

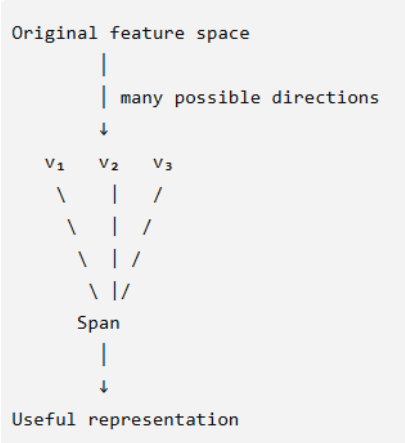
For example, two non-collinear vectors in 2D span the entire plane, while collinear vectors span only a line. In 3D, three non-coplanar vectors span the entire space, and coplanar vectors span a plane.

For vectors $v_1, v_2, \dots, v_n$:

$$\boxed{\text{Span}\{v_1, v_2, \dots, v_n\} = \{a_1 v_1 + a_2 v_2 + \cdots + a_n v_n\}}$$

where $a_1, a_2, \dots, a_n$ can be any scalars.

Suppose: $v = (2, 1)$, Its span is: $\text{Span}\{(2, 1)\}$
This means: $a(2, 1)$, where a can be any real number.



Example 1: Span of Two Vectors in $\mathbb{R}^2$

Consider two vectors $v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$

The span of $\{v_1, v_2\}$ is the set of all vectors that can be written as:

$$c_1 v_1 + c_2 v_2 = c_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + c_2 \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

This can be written as:

$$\text{Span}\{v_1, v_2\} = \left\{ \begin{pmatrix} c_1 + 3c_2 \\ 2c_1 + 4c_2 \end{pmatrix} \middle| c_1 \middle| c_2 \in \mathbb{R} \right\}$$

Since v_1 and v_2 are not collinear (they are not multiples of each other), they span the entire $\mathbb{R}^2$ plane.

Example 2: Span of Three Vectors in $\mathbb{R}^3$

Consider three vectors $u_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, u_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and $u_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

The span of $\{u_1, u_2, u_3\}$ is the set of all vectors that can be written as:

$$c_1 u_1 + c_2 u_2 + c_3 u_3 = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + c_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

This can be written as:

$$\text{Span}\{u_1, u_2, u_3\} = \left\{ \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} \middle| c_1 \middle| c_2 \middle| c_3 \in \mathbb{R} \right\}$$

Since $u_1, u_2,$ and u_3 are linearly independent, they span the entire $\mathbb{R}^3$ space.

A linear combination is one vector obtained by multiplying given vectors by scalars and adding them, whereas the linear span is the complete set of all vectors that can be obtained through such linear combinations.

USE OF SPAN IN AI:

AI represents data as vectors. An image might have thousands of pixel values, but many of these features may be related to each other. So a complicated feature vector can be represented as a combination of smaller number of important feature vectors. That smaller set of directions defines a span.

```
Image
 ↓
Thousands of pixel values
 ↓
Features
 ↓
AI model
```